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Progress of the R&D programme to develop a metal foil pump for DEMO

Stefan Hanke[a,*], Christian Day[a], Thomas Giegerich[a], Juri Igitkhanov[a], Yannick Kathage[a], Xueli Luo[a], Stylianos Varoutis[a], Alejandro Vazquez Cortes[a], Thomas Härtl[b], Andrei Busniuk[c], Alexander Livshits[c], Stefan Merli[d], Andreas Schulz[d], Matthias Walker[d], Klaus Baumgärtner[e], Jens Hofmann[e]

[a] Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany
[b] EUROfusion Consortium, PPPT Department, Garching, Germany
[c] Bonch-Bruevich Saint-Petersburg State University of Telecommunications, Saint-Petersburg, Russia
[d] University of Stuttgart, Stuttgart, Germany
[e] Muegge GmbH, Reichelsheim, Germany

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ABSTRACT

DEMO requires a novel fuel cycle architecture with Direct Internal Recycling in order to drastically minimize the tritium inventory. One central element of this architecture is the separation (and compression) of hydrogenic gas from the exhaust gas stream, close to the divertor, which can immediately be recycled to feed the matter injection systems. The hydrogen-selective separation will be achieved by the metal foil pump, a novel pump type relying on the effect of superpermeation, which is a first-of-its-kind application for the conditions to be matched for DEMO. To achieve a convincing performance, each hydrogen particle has to undergo successfully a series of processes before it is separated and pumped. To support the design of such a pump, a comprehensive R&D programme has been started which is currently being focused on (i) the plasma source, (ii) the transport of the suprathermal hydrogen, and (iii) the material and surface aspects of the foil. The experimental and modelling tools are described, examples of recent results are presented and a first prediction on the performance is derived.

1. Introduction

One of the challenges in the development of the DEMO fuel cycle is given by the need to avoid excessive tritium inventories.

The Direct Internal Recycling (DIR) concept and the KALPUREX-process [1] tackle the issue via bypassing the tritium plant with a low-pressure, selective hydrogen pump in proximity of the torus and thereby decreasing the tritium inventory significantly [2]. This pump exploits the effect of superpermeation [3] to separate a large fraction of the hydrogen (tritium, deuterium and small amounts of protium) from the torus exhaust and feed it back to the fueling systems. Characteristic for superpermeation is that the permeation can work against pressure gradients. Superpermeation can be achieved with the production of suprathermal hydrogen particles, which impinge on a thin metallic membrane, giving the pump its name: Metal Foil Pump (MFP). The feasibility of hydrogen separation in a MFP has a strong impact on plant and tokamak architecture. It has, therefore, become one of the 8 key design integration issues which are being addressed in the European DEMO Programme with special focus until 2020 [4].

2. Necessary knowledge basis for a metal foil pump

The key to superpermeation can be found in the surface-limitation [3] of the hydrogen permeation through a metal foil and a supply of suprathermal hydrogen upstream. It is illustrated in Fig. 1.

Suprathermal means energized above the thermal energy of the particle in the gas phase. This excitation is indicated by step ① and it can be carried out by an external power source. The suprathermal hydrogen particles with a non-uniform energy distribution can impinge on said foil as indicated in step ②. Step ③ represents the surface interaction processes of these suprathermal particles. These processes strongly depend on the state of the surface (the surface barrier ideally originates from an impurity monolayer [3,5]). Once the hydrogen enters the bulk, it diffuses back and forth through the lattice due to a random walk process in step ④. Eventually, it is released into the gas phase after finding its way to a surface and recombining with another hydrogen atom in step ⑤. This can happen on both sides of the membrane. The probability for downstream release depends on the ratio of the energy barriers on both sides. This probability is always several orders of

* Corresponding author.
E-mail address: stefan.hanke@kit.edu (S. Hanke).

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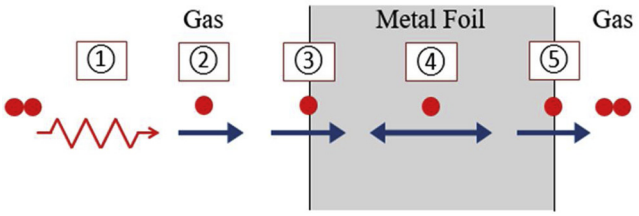


Fig. 1. Sequential steps of a hydrogen particle in the superpermeation process.

magnitude larger than the permeation probability in the diffusion limited regime for low implantation depths.

Once the hydrogen molecule is released on the downstream side as a thermal molecule, it is by several orders of magnitude less likely to be absorbed into the lattice again as compared to the initial suprathermal hydrogen [6,7].

This yields an effective pumping, even possible against large pressure gradients [8]. Superpermeation only occurs for hydrogen and its isotopes [9], making it the perfect tool for their separation from impurities like helium or plasma enhancement gases.

To come up with a workable design of the MFP, each of the 5 steps described above has to be fully understood for DEMO-relevant conditions so that the operational range of the pump is maximized. Superpermeation is being experimentally investigated at KIT in the new HERMES*plus* facility, which is equipped with a filament and a collisional plasma source to produce suprathermal hydrogen for a tubular metal foil of 30 cm^2 in a stainless steel plasma chamber. Niobium (Nb) is being tested as a metal foil material first and a vanadium (V) membrane module will be tested in the future as well. A sketch of the facility is shown in Fig. 2. In the following, the individual steps of the process introduced in Fig. 1 are described with reference to our recent results and plans.

2.1. Suprathermal hydrogen production

The efficiency of the desired permeation process relies heavily on the effectiveness of the suprathermal hydrogen supply. The source of this supply has to be researched itself as well as the interplay with the metal foil. There are multiple methods for producing suprathermal hydrogen suitable for the above described process. One of those is an incandescent filament as we were also able to confirm in previous experiments. HERMES*plus* contains a filament only to have a baseline for comparison with literature data. For a number of reasons, filaments are no candidate for a MFP in DEMO. Mainly, the pressure expected for a MFP downstream the divertor is too high for sufficient suprathermal hydrogen production with a filament. Furthermore, the filament at the necessary high operational temperatures (of the order of 2000 K) is prone to lifetime issues and sensitive to pressure jumps that may occur

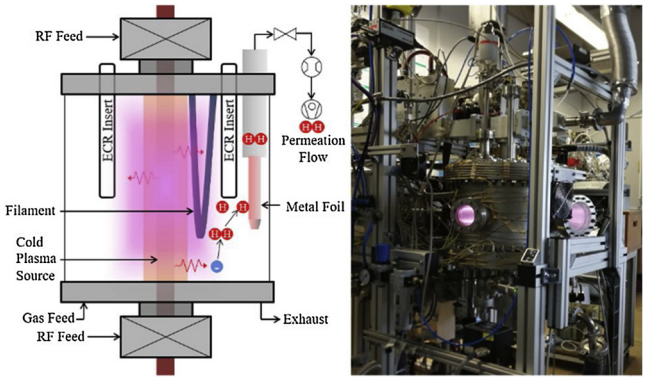


Fig. 2. Sketch and photo of the HERMES *plus* facility. The permeation flow through the metal foil can be measured.

in the course of safety events. To overcome these issues, we aim to develop and qualify a new microwave plasma source for generation of suprathermal hydrogen - if needed with extended operational pressure range by Electron-Cyclotron-Resonance (ECR) plasma.

Currently under investigation at our experimental setup is the collisional plasma source *Duo-Plasmaline* (developed and manufactured by Muegge GmbH, Germany) which injects a 2.45 GHz microwave into the vacuum chamber [10]. The electric field component of the wave accelerates charged particles present in the chamber. Here, this is mainly true for electrons as ions are too heavy to exhibit large displacements by a wave of this frequency. Hence, their temperature stays in equilibrium with the neutral gas while electrons are accelerated to higher velocities. Only if the frequency of elastic collisions between neutral gas particles and electrons during their displacement is within a certain regime, a net energy transfer to the gas is possible and plasma can ignite. These elastic collisions act as a channel of energy conservation for the electrons as their movement is redirected to gain velocity components non-parallel to the electric field and the total velocity can increase up to values which allow inelastic collisions [11]. Those can be, for example, dissociation in hydrogen gas or ionization, leading to an electron avalanche and the ignition of plasma. The evolution of suprathermalized hydrogen upon electron impact is depicted in the sketch in Fig. 2.

Collisional plasmas only ignite in certain pressure ranges. In HERMES*plus* we found a strong correlation between this pressure limit and the power input with operation being possible down to well below 10 Pa. In view of the fact that the separation effect of the MFP is only needed during the divertor pressure regime in the plasma burn phase, the achieved pressure limit is already in a very promising range. For the significantly smaller throughputs during dwell, no separation is needed, and the vacuum pumps installed further downstream the pumping line can be utilized. In case future R&D will not be sufficiently successful to further extend the pressure limit of the collisional plasma in magnetic field, there is a fallback solution to extend the operation range to lower pressures by establishing the ECR condition [12].

Obviously, the external magnetic field at the location of the MFP in DEMO has an influence on the performance of the plasma source. This is why a separate experiment is currently being set up to study such impact, see section 3. Satisfying the ECR condition at the typical magnetic field strengths of the order of 1 T present at the location of the MFP would require going to higher frequencies (> 2.45 GHz) of the plasma source.

A key characteristic of the suprathermal hydrogen source in which we are interested in, is its effectiveness in attaining higher permeation flows. The yield of suprathermal particles generated in the source can be investigated experimentally in HERMES*plus*. A two-channel broadband spectroscope (Plasus GmbH, 200–1100 nm range, 1.5 nm spatial resolution) is under commissioning to study the plasma composition. One of the spectroscope optics will be fixed in position in vacuum, covering the line of sight along the *Duo-Plasmaline* while the other is adjustable ex-vacuum to cover different radial distances from the plasma source. The optical actinometry method [13] allows the determination of atomic hydrogen concentrations via the relative peak intensity of the characteristic H_α, H_β and H_γ lines to the peak intensities of an actinometer gas (for example argon) at an exactly known concentration. A Langmuir probe would yield information on electron density and temperature at the same radial distance from the plasma source as has the metal foil. These measurements serve the purpose of validating a COMSOL Multiphysics simulation of the plasma, which is currently under development. Previous experimental investigation of this plasma source indicates that the electron temperatures should be around 3 eV with electron densities around $10^{18}\,m^{-3}$ [10].

Surface recombination has a major impact on the composition of the plasma [14]. In our setup, most of the plasma-wall interface surface belongs to the stainless steel vessel. A fully fleshed out MFP design might, however, feature other configurations, in which atom-atom recombination on the metal foil material is more dominant and yields a

different atomic hydrogen concentration in plasma. Literature data on atomic sticking coefficients differ greatly [6,15,16], as do the atom-atom recombination rates. Once simulation and experiment have been calibrated to each other, the simulation can be used to make predictions for a large-membrane setup.

2.2. Suprathermal hydrogen transport

Obviously, the metal foil pump can work only if the suprathermal particles reach the metal foil surface. This means the transport of such hydrogen species to the metal foil has to be realized with minimum losses. This defines the geometric configuration of the plasma source relative to the metal foil. Occurrence of volume and surface recombination of suprathermal particles and loss rates of excited states have to be taken into account when integrating the plasma source and metal foil towards a workable design of a pump for the available space in the lower ports of DEMO. In modelling the relevant processes, the full range of gas collisionality has to be taken into account. Consequently, two approaches are implemented for modeling the flow domain between plasma source and metal foil: The Test Particle Monte Carlo (here we use the code ProVak3D [17]) and the Direct Simulation Monte Carlo method (here we use the code SPARTA [18]).

Fig. 3 shows an initial conceptual design of a MFP with a tubular shape metal foil around the linear plasma source, which forms the central axis. Large arrows indicate the flow of the gas, which enters from one side and leaves it via the foil (hydrogenic species) or through the other end of the pump (impurities and non-permeating hydrogenic species) [19].

Particle transport simulations have been performed with ProVak3D for a 2 m long pump at a foil diameter of 0.5 m and an assumed plasma core diameter of 0.1 m. As an example how these simulations can be translated to derive valuable information for the designers, Fig. 4 compares two cases. One of which shows moderate performance of the foil, when the probability of a single suprathermal particle to permeate through the foil upon impingement on it is 5 %. The other curve assumes a higher permeation probability of a single suprathermal particle of 20 %. The probability of de-excitation into a ground-state particle upon foil impingement is 15 % and 0 % respectively, which leaves a reflection probability of 80 % in both cases. The assumed plasma performance is not varied in the given example.

Along its axis, the model is separated into 40 rings for which individual pumping probabilities of the foil are derived as the ratio of number of particles permeating through the ring surface vs. the number of particles entering the model through the inlet surface. Adding the individual ring's pumping probabilities together yields the total pumping probability of the foil, which can then be used to calculate a pumping speed using the pump's cross-section and the thermal velocity of the particles. The total pumping speed composes of the foil pumping and the rough-pumping via the outlet surface. Overall, it is found that the second half of the pump is contributing less to its performance, which comes from conductance limitation within the tube. By integration of the curves along the axis, one obtains a total pumping speed of 23.9 m^3/s for the high performance case and a value of 13.4 m^3/s for the low permeation case. This illustrates that the pumping

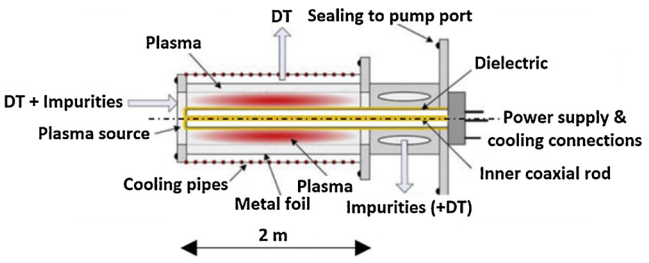


Fig. 3. Sketch of an initial MFP design.

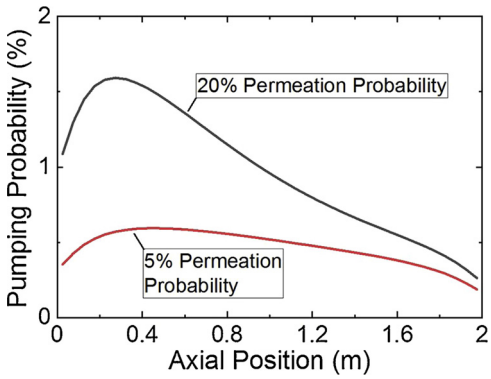


Fig. 4. Calculated pumping performance along the axis of the pump.

speed scales clearly less than linearly with the metal foil performance. It holds even more pronounced for the separation factor, which is defined as the ratio of the pumping probability of the foil to the total pumping probability of the pump, which includes pumping via rough-pumps that are considered as a sticking coefficient of the model end wall. The separation factor changes only by 17 % (from 82 to 96 %) for the shown change of the permeation probability by factor 4. This is due to the fact that multiple hits of energetic particles can compensate if the first hit does not lead to successful permeation immediately.

2.3. Suprathermal hydrogen interaction with surface barrier

The definition of the upstream surface state and its energy barrier to hydrogen absorption is one of the most important tasks in the roadmap to a MFP development. Superpermeation works only in the surface limited regime and the barrier determines the main characteristics of superpermeable membranes [3,20]. In order to operate a MFP with predictive surface manipulation, the dominant surface processes have to be identified and the effect of the cold plasma on the foil has to be understood.

The maximum achievable superpermeation flux was evaluated for Nb [20], Pd [21] and Fe [22] experimentally measuring the value of the surface barrier. Still, the physio-chemical origin of the barrier remains not quite clear up to now. Ab-initio molecular dynamics modelling of the surface will be carried out in order to obtain a clear picture of how the environment affects the energy barrier and investigate relations between surface state on the one hand, and recombination and sticking coefficients on the other hand. For this reason, supporting surface analysis of the membrane materials, both ex-situ and in-situ, is a necessary activity to be included in the R&D programme towards a MFP. Ex-situ techniques are XRD (X-Ray Diffraction) characterization for crystal lattice modifications due to impurities, SEM-EDS (Scanning Electron Microscopy-Energy Dispersive X-ray Spectroscopy) for grain size and composition of impurity layer, and AES (Auger Electron Spectroscopy) with sputtering for an in-depth composition analysis. In-situ analysis is carried out by thermal desorption and a mass spectrometer.

Each side of the foil is in contact with different environments. In particular, the plasma on the upstream side can potentially create defects if it contains high energy ions or deposit undesired impurities on the metal foil surface, and thus may change its characteristics over time [23–25]. However, the currently investigated microwave plasma source generates a cold plasma, which typically features insufficient ion energies and plasma potentials for substantial physical sputtering [26]. It should be noted that even for slightly unfavorable asymmetric energy barriers on up- and downstream side, the plasma driven permeation still works and still is, by orders of magnitude, stronger than any gas driven permeation at similar conditions.

The material permeation characteristics of the membrane are key. Much research has been carried out regarding this matter. Waelbrock

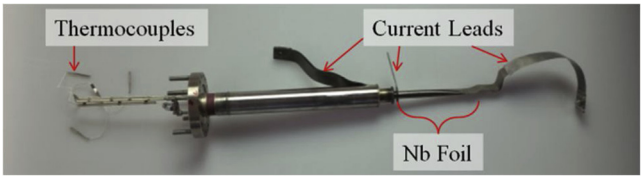


Fig. 5. Metal foil module (Nb, 0.1 mm thick), manufactured by St. Petersburg State University of Telecommunication.

et al. in Jülich focused mainly on iron coated membranes with very encouraging results [22,24] and the group of Livshits put its focus on group 5 metals with a natural O monolayer that provides the energetic barrier for molecular absorption and desorption [3,20]. We are following the latter course, as this provides us with the opportunity of in-situ recovery of the monolayer, in case there is any degradation over time [27,28]. The most promising of the group 5 metals appear to be Nb and V as they feature the highest permeation fluxes, exhibit no hydride formation or blistering at acceptable temperatures and have good mechanic properties as well as allow formation of appropriate surface barriers due to impurity adsorption. Membrane modules of both materials are available for experiments in HERMES*plus* already. The Nb membrane module is shown in Fig. 5. It is made in the form of a tube of 1 cm in diameter and 10 cm in length with a wall thickness of 0.1 mm. One end of the tubular membrane is merged and welded to a current lead and the other end is welded to a stainless steel holder via which the membrane sample is fixed in the plasma chamber (compare Fig. 2). The holder has a ceramic break that electrically isolates the membrane sample from the chamber wall. This provides the possibility to apply an electrical bias on the membrane sample relative to the plasma and therefore to vary the energy of incident ions. The membrane sample can be resistively heated to any desirable temperature (up to the melting point). A special design of the current leads shown in Fig. 5 ensures a uniform temperature of the metal foil. The temperature is measured via a thermocouple and a ratio-pyrometer. The membrane was produced from cold rolled niobium foil without any special surface treatment except solvent rinsing. AES analysis shows oxygen and carbon contamination. After metal foil heating in vacuum, the surface impurities dissolves in the bulk and a monolayer of oxygen remained on the surface.

2.4. Role of hydrogen transport through the bulk

Operation of a MFP will take place in surface-limited regime. To ensure this, the diffusion of hydrogen in the metal bulk should pose no limit to the permeation flow. In this case, the permeation flux can be nearly half of the absorption flux (for symmetric membranes), or almost equal to it (higher inlet than outlet surface barrier [3]). Then, the performance of the foil is temperature independent as long as it stays in the appropriate temperature window described below. Since suprathermal hydrogen species are absorbed regardless of temperature, a fine temperature control of the foil is not necessary.

Bulk characterization is an important step towards fully understanding the permeation process, for instance, in the role of diffusion of solute oxygen for the replenishment of the monolayer or the mechanical properties of the foil. The bulk characteristics of the metal lattice define its diffusivity. Therefore, their modification, e.g. by hydride formation, will have an effect on the permeation, besides the mechanical issues that hydrides can cause. In order to avoid hydride formation, the temperature of the membrane should be higher than 171 °C in the case of a V or Nb membrane [29,30]. Furthermore, the material defects of a real metal lattice (crystallite planes) at low temperatures also play a role in the hydrogen permeability. On the other hand, the high temperatures increase the diffusivity of dissolved O atoms, enhancing the monolayer self-healing feature of the membrane [27]. The hydrogen solubility in Nb or V decreases with increasing temperature because of its negative enthalpy of solution [31–33]. Thus, an increase in the temperature of the metal leads to a decrease in the concentration of hydrogen in the metal lattice under steady-state conditions. This also occurs during the interaction of the metal with non-equilibrium hydrogen (permeation driven by suprathermal hydrogen). This helps to maintain low hydrogen concentrations in the metal, since even without hydride formation, high concentrations can pin dislocations and cause embrittlement. Therefore, in group 5 metals, increase of temperature and the subsequent reduction of the solubility would also lead to a reduction in the maximum tritium inventory. From this point of view, membrane temperature should be kept as high as possible, without shifting to the diffusion-limited regime. Therefore, the foil temperature should be kept below a certain value that ensures staying in the surface-limited regime.

In summary, one of the aims of the HERMES*plus* experimental campaigns is the search for an optimal operational metal foil temperature that holds a balance between low tritium inventory and high performance and reliability.

2.5. Thermal hydrogen absorption & desorption from metallic surfaces

The rates of recombinative desorption of solute hydrogen atoms and dissociative absorption of hydrogen molecules are radically decreased by the energy barrier created by the nonmetal impurities on the surfaces of the metal foil. Based on the presence of vacancies in the surface coverage with impurities, solute hydrogen is released into the gas phase as thermal molecules. The molecular absorption depends on the coverage and the membrane temperature. Therefore, the absorption-desorption kinetics may be modified via those two features [6]. As a result, the concentration of hydrogen (tritium) during permeation driven by suprathermal hydrogen can be modified via the membrane temperature. On the other hand, the suprathermal hydrogen absorption probability does not depend on the membrane temperature in the pursued operation regime and stays larger than molecular absorption by a few orders of magnitude at any temperature [3]. Information on the absorption and desorption probabilities will be gained in HERMES*plus* experiments with plasma switched off [8]. Thus, conditions for classical gas driven permeation are provided, in which the permeated flux (significantly lower than in plasma driven permeation conditions) is mainly given by absorption and desorption rates for the different surface states and membrane temperatures. Since the process of molecular driven permeation through Nb and V membranes are surface limited, these measurements are capable of giving the information on absorption-desorption kinetics (surface barrier) at different states of surface and temperature.

2.6. Integration aspects

In HERMES*plus*, permeation fluxes of up to 1 Pa·m^3/s per m^2 metal foil area have been demonstrated so far [34]. Though further improvements of this value are expected during the ongoing pump development process, this value allows already now the dimensioning of MFPs in DEMO-scale. Several of these pumps, for which the tubular design as shown in Fig. 3 is currently foreseen, will be combined in so-called pump casks to achieve the desired overall pumping performance. These casks will be installed in the DEMO lower ports, using remote maintenance operations [19]. It is obvious that the development of pumps that have to fit into pump casks which again have to fit into the ports is a highly integrated task.

Another challenge in MFP development and integration results from the operation conditions: High foil temperatures have to be handled and alternating currents for foil heating and plasma generation have to be supplied, considering thermal and/or mechanical limitations by the pump's internal- (support structure) and external structures (pump cask and pump port).

Altogether, the design development process of MFPs and its

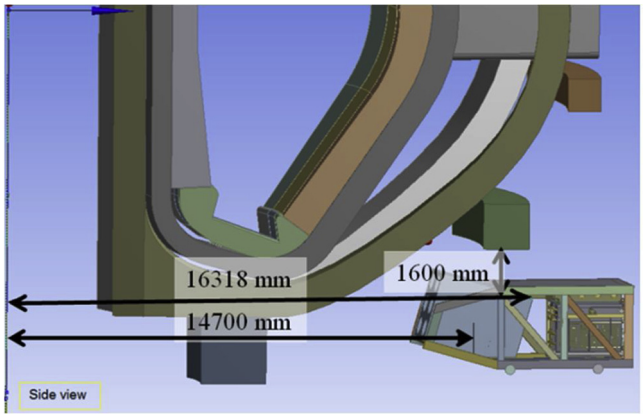


Fig. 6. Illustration of the potential pump location relative to the coils. The grey box denotes a MFP unit comprising of six MFP cylindrical arrangements.

integration into the DEMO design is an iterative task that combines thermo-mechanical simulations, load calculations, radiation and magnetic field simulations. This work has been started in dedicated tasks in the EUROfusion Work package on Tritium-Matter Injection and Vacuum.

3. Main risks for a metal foil pump

At the start of the development of a fully functioning MFP for application in DEMO, all possible risks for the system were identified and appropriate measures taken to mitigate or find solutions. This has pointed to four main aspects which are now under investigation.

One aspect comes from the MFP's proximity to the torus and therefore to the superconducting coils of the tokamak. Fig. 6 illustrates the preliminary pump location around 2 m below the PF 6 coil, where magnetic field strengths of the order of 1 T are present.

Such strong magnetic fields will have an effect on the MFP plasma and cause its electrical conductivity to become anisotropic. Depending on the field orientation relative to plasma, this can lead to the compression of the plasma, increased losses to the wall, changes in charge densities and localized heat sinks. This issue will be investigated in a series of COMSOL Multiphysics simulations, which will be validated in large-scale experiments with a *Plasmaline* (provided by Muegge GmbH) and an external magnetic field as shown in Fig. 7. A current of up to 2 kA can be fed into the surrounding water-cooled coils to produce a magnetic flux density of up to 250 m T within a few hundred ms. While this does not fully cover the range of magnitude expected in DEMO, the results can be reliably extrapolated as no further effects shall occur beyond the critical field strength of 87.5 m T (ECR condition). A

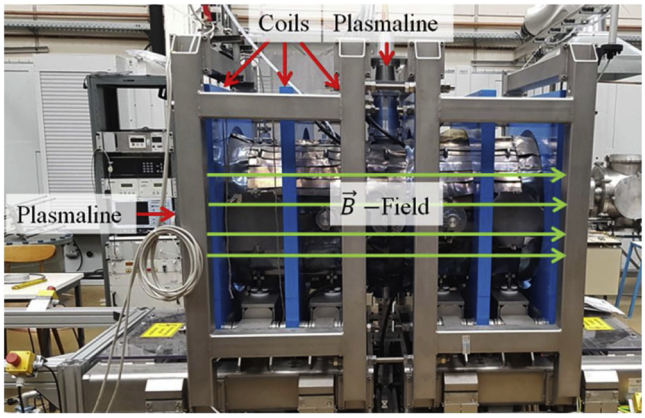


Fig. 7. Experimental set-up to study the influence of external magnetic fields, located at University Stuttgart.

Plasmaline with a one-sided microwave feed of up to 6 kW power input will first be immersed parallel and then perpendicular to the magnetic field and tests for different microwave powers, pressures and magnetic field strengths will be carried out. Diagnostics include optical spectroscopy next to observation of standard experimental parameters and possibly electron density and temperature via a Langmuir probe.

However, a partial shielding of the magnetic field might allow utilizing it to operate ECR plasma without installation of further auxiliary systems. The option of magnetic shielding of the pump duct is currently examined with a Finite Element Method Magnetics simulation.

The lack of reliability of the permeation fluxes that has been noticed by several different authors [6,23,33], despite the same incident flux and membrane conditions, is the second major risk. This is probably due to the instability of the surface state with time under influence of external agents. The aspect of long term stability of the metal foil surface will be addressed in a repeatability test, which measures permeation in distinct standard operation condition, allowing a comparison of the results over an extended period of time.

The third aspect, also relevant for the long-time performance of the MFP, is the capability of healing of the upstream surface barrier. At which time scales healing is necessary has to be investigated in parametric experiments. If there is a need, the replenishment of this barrier should be performable in-situ in order to relax maintenance requirements. Diverse options are suggested in literature [6,28,33]. The most convenient, in the case of loss of coverage, is the self-healing of a natural growing O monolayer by internally dissolved O atoms in the lattice. This, however, has the disadvantage that eventually the metal foil will run out of O and then the foil would have to be substituted. This could be avoided by exposing the metal foil to a sub Pa O_2 pressure, so O is absorbed and the monolayer regrown. An application of this process during dwell time would avoid the formation of heavy water. This surface healing technique will be tested in HERME*Splus*. In the case of unwanted deposition of impurities on the surface, leading to more than a monolayer coverage, and the consequential hindering of atomic sticking, one can use high temperatures or sputtering by application of a membrane bias [3] in order to re-condition the foil in-situ. Both approaches will be tested in HERME*Splus*.

Finally, the permeation in a MFP for DEMO will have multi-isotopic features, making the interaction among protium, deuterium and tritium something worth looking into. It is the aim to find operational regimes under which the integral isotope effect, i.e. over all the five steps of Fig. 1, is minimal, so that the cut of the permeating gas that can be utilized for DIR is maximized. This also poses a direct requirement on the performance of the Isotope Rebalancing Unit in the inner loop of the fuel cycle. We will further investigate permeation characteristics for H_2 and D_2 in our setup, as well as for heterogeneous HD molecules. A test with tritium for design validation purposes is foreseen at a later stage in time, but still as part of the conceptual design phase of DEMO.

4. Conclusion

Within the next years, a fully functioning MFP for DEMO application will be developed in a joint effort with university, research institutions and industry. A R&D programme has been developed which is aimed to address possible showstoppers and risks. This programme includes experiments and supporting theoretical analyses.

The optimization of permeation efficiency is one of the main objectives in the MFP development, next to repeatability, reliability and low maintenance.

CRediT authorship contribution statement

Stefan Hanke: Writing - original draft, Conceptualization, Project administration. **Christian Day:** Supervision, Funding acquisition, Project administration, Writing - review & editing. **Thomas Giegerich:** Visualization, Writing - review & editing. **Juri Igitkhanov:** Writing -

review & editing. **Yannick Kathage:** Writing - original draft, Visualization, Investigation, Writing - review & editing. **Xueli Luo:** Software. **Stylianos Varoutis:** Software. **Alejandro Vazquez Cortes:** Writing - original draft, Investigation, Writing - review & editing. **Thomas Härtl:** Funding acquisition. **Andrei Busniuk:** Writing - review & editing. **Alexander Livshits:** Writing - review & editing. **Stefan Merli:** Software. **Andreas Schulz:** Writing - review & editing. **Matthias Walker:** Writing - review & editing. **Klaus Baumgärtner:** Resources. **Jens Hofmann:** Resources.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

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